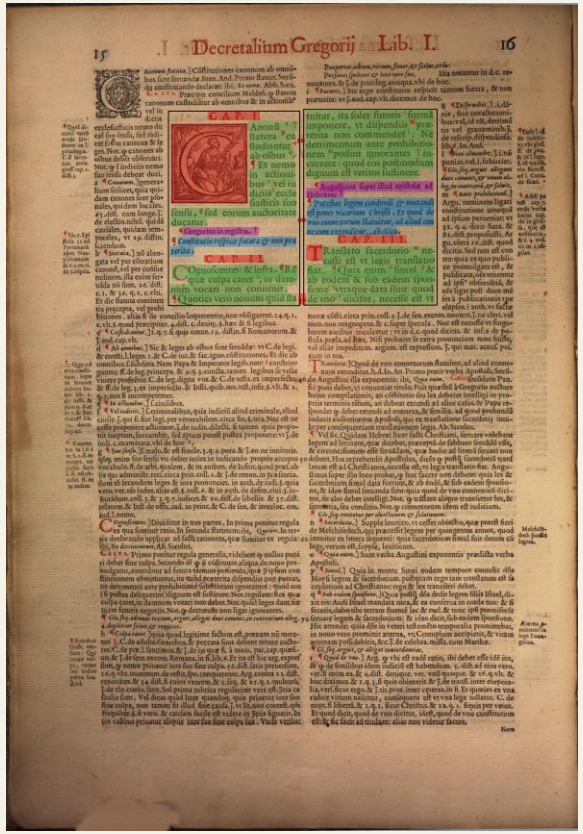
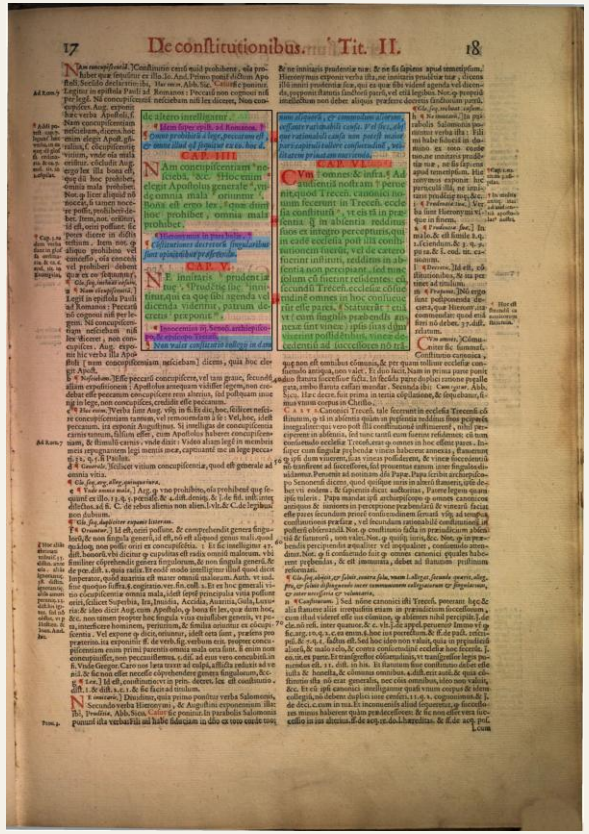
The two-page task: 131 / 131 correct

Rome, 1582 · Decretals · MDZ bsb10144443 · scans 40 and 41





text · summarium · inscriptio · other

A complete, reproducible submission

Fetch images + hOCR

Find the two central blocks

Assign the four requested roles

Export overlays + evaluation

PERSONALLY CHECKED AGAINST BOTH PAGES

OpenCV + NumPy · no training required

A small rule system is sufficient for this edition

The visual distinctions stated in the brief become explicit measurements.

Red ink	→ other
Slanted type	→ summarium
Large black type	→ text
Remaining small type	→ inscriptio

Reference → program

	text	summ.	inscr.	other	outside
text	**60**	0	0	0	0
summ.	0	**16**	0	0	0
inscr.	0	0	**7**	0	0
other	0	0	0	**6**	0
outside	0	0	0	0	**42**

Also labelled: gutter numbers + initial

131 checked labels · all diagonal

Adaptive thresholding · split gutter-crossing OCR lines · grow each block to its chapter head

Across editions, use learned page and line models

D-FINE finds regions. PERO reads inside them. The neural tagger assigns roles.

D-FINE

region detection

→

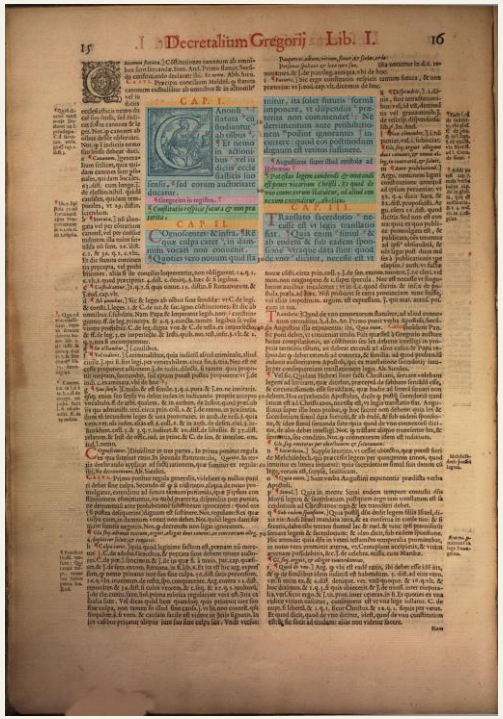
PERO

lines + recognition

→

Neural v3

region + line roles



120 pages

Region dataset
100 training + 20 evaluation

120 pages

Role dataset
100 training + 20 evaluation

Line image + OCR characters + position + page context

The trained systems have measurable results

The two evaluations answer different questions: finding regions and assigning roles.

REGIONS · D-FINE

0.862

region F1 at IoU ≥ 0.50

20 evaluation pages from 20 other books
206 matched · 32 extra · 34 missed regions

ROLE LABELS · NEURAL v3



0.945 macro-F1 · 20 evaluation pages
14 label errors among 5,065 supplied lines

Training priorities: small source labels, chapter apparatus and precise region boundaries.

Six days: a complete experimental environment

LLM agents made this breadth of implementation and testing possible.

100	26k+	200+	3
volumes inventoried	source lines	source files	neural versions

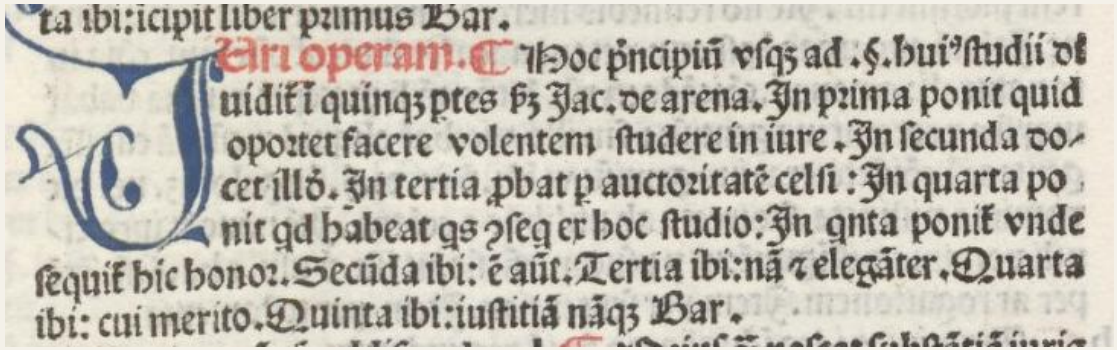
Acquire → annotate → train → evaluate → inspect → revise

Claude + Codex: implementation, debugging and experiments
ChatGPT: training annotations
My work: research direction, decisions and personal checking

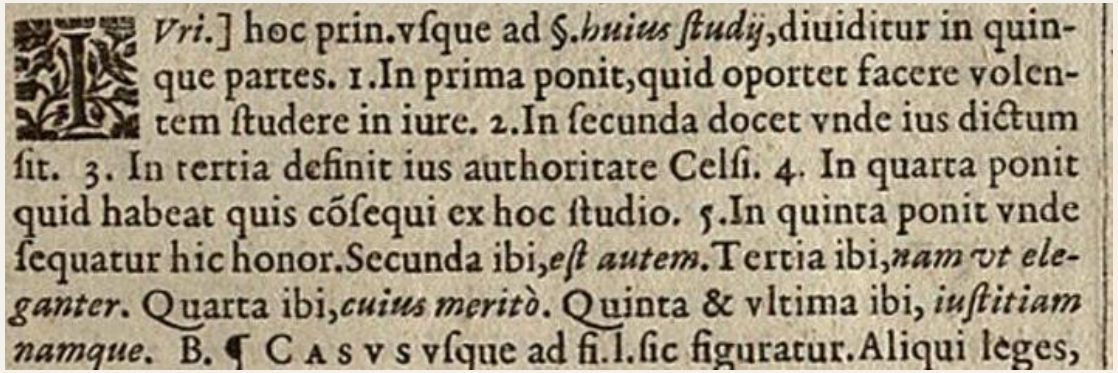
The same divisio; a different attribution

D. 1.1.1pr · the supplied witnesses, Venice 1490 and Lyon 1627

1490


ta ibi:icipit liber primus Bar.
Ari operam. ¶ Hoc ꝓncipiũ vſqꝫ ad .§.huiꝰ ſtudii dꝫ
uidit i quinqꝫ ptes ꝗꝫ Jac. de arena. In prima ponit quid
oportet facere volentem ſtudere in iure. In ſecunda do-
cet illꝺ. In tertia ꝓbat ꝓ auctoritatẽ celſi: In quarta po
nit qd habeat qs ꝙſeq er hoc ſtudio: In qnta ponit vnde
ſequit hic honor. Secũda ibi: ẽ aũt. Tertia ibi: nã ꝛ elegãter. Quarta
ibi: cui merito. Quinta ibi: iuſtitiã nãqꝫ Bar.

1627


Vri.] hoc prin. vſque ad §. *huius ſtudij*, diuiditur in quin-
que partes. 1. In prima ponit, quid oportet facere volen-
tem ſtudere in iure. 2. In ſecunda docet vnde ius dictum
ſit. 3. In tertia definit ius authoritate Celſi. 4. In quarta ponit
quid habeat quis cõſequi ex hoc ſtudio. 5. In quinta ponit vnde
ſequatur hic honor. Secunda ibi, *eſt autem*. Tertia ibi, *nam vt ele-
ganter*. Quarta ibi, *cuius meritò*. Quinta & vltima ibi, *iuſtitiam
namque*. B. ¶ Casvs vſque ad fi. l. ſic figuratur. Aliqui leges,

Jac. de arena … Bar.

Jacobus de Arena + Bartolus

B.

Bartolus

The historical event to detect: Jacobus de Arena disappears from the attribution.

Retrieve the legal locus, then recover its paratext

An edition-independent anchor connects different page layouts.

1 Anchor

D. 1.1.1pr
Book / title / fragment / incipit

→

2 Retrieve

OCR character n-grams
RapidFuzz: fuzzy text matching

→

3 Attach

Layout + lemma + sequence
Find the neighbouring divisio

→

4 Assemble

Join lines into a whole unit
Keep image coordinates

Unit key edition × legal locus × paratext type × occurrence

Compare wording and historical content separately

Normalize for matching; preserve the evidence needed to detect change.

Diplomatic layer

Raw OCR → checked reading → image link



Matching layer

Abbreviation alternatives · long-s / u-v
CollateX alignment · candidate similarity

Content layer

Names · legal citations · negation
Conditions · exceptions · attribution

Bar. ↔ B. → Bartolus

Jac. de arena → Jacobus de Arena

Cluster aligned variants; keep every deletion of a name or qualification visible.

Make the change an explicit, inspectable record

One aligned unit; two resolved authority sets; one deletion event.

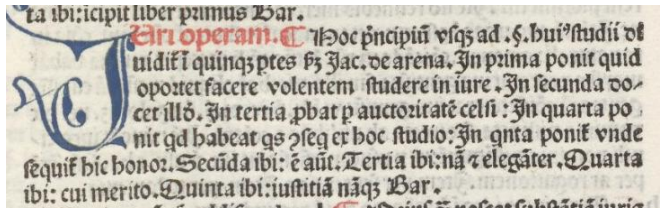
1490		1627
Jacobus de Arena Printed: Jac. de arena	— ×	ATTRIBUTION REMOVED
Bartolus Printed: Bar.	→	**Bartolus** Printed: B.

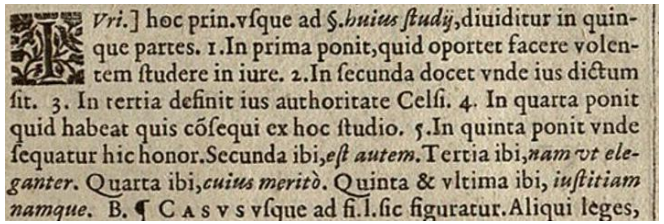
Event: D.1.1.1pr / divisio / Jacobus de Arena / present in 1490 → absent in 1627

Return evidence to historians—and patterns to research

Proposed comparison desk: link each event to both witnesses.

D. 1.1.1pr / divisio / attribution removed





1490: Jacobus de Arena; Bartolus
1627: Bartolus

CONFIRM | CORRECT | EXPORT EVIDENCE

Main risk

OCR loss masquerades as historical deletion.

Inspect the complete source unit; re-recognise or transcribe the disputed span.

Research payoff: where, when and in which editorial families did attribution disappear?

A 360-page pilot across six editions

1 December 2026 → 31 May 2027 · matched legal loci + contrasting page structures

Work	Edition	Purpose	Pages
Digest	Venice 1490	Gothic + gloss	**60**
Digest	Lyon 1569	Roman + gloss	**60**
Digest	Venice 1606	Roman + Greek notes	**60**
Digest	Lyon 1627	Roman + dense apparatus	**60**
Decretals	Venice 1494	Gothic + gloss	**60**
Decretals	Rome 1582	Roman + gloss	**60**

40 Digest loci × 4 editions
20 Decretals loci × 2 editions

240 train / 60 development / 60 test
Freeze test pages before iteration.

Historians define the units and the distinctions

Annotation concentrates expertise on decisions with historical consequences.

Page structure

Regions · line order · continuation
Main text / gloss / additions / paratext

Unit identity

Legal locus · lemma · paratext type
Where the unit begins and ends

Text and meaning

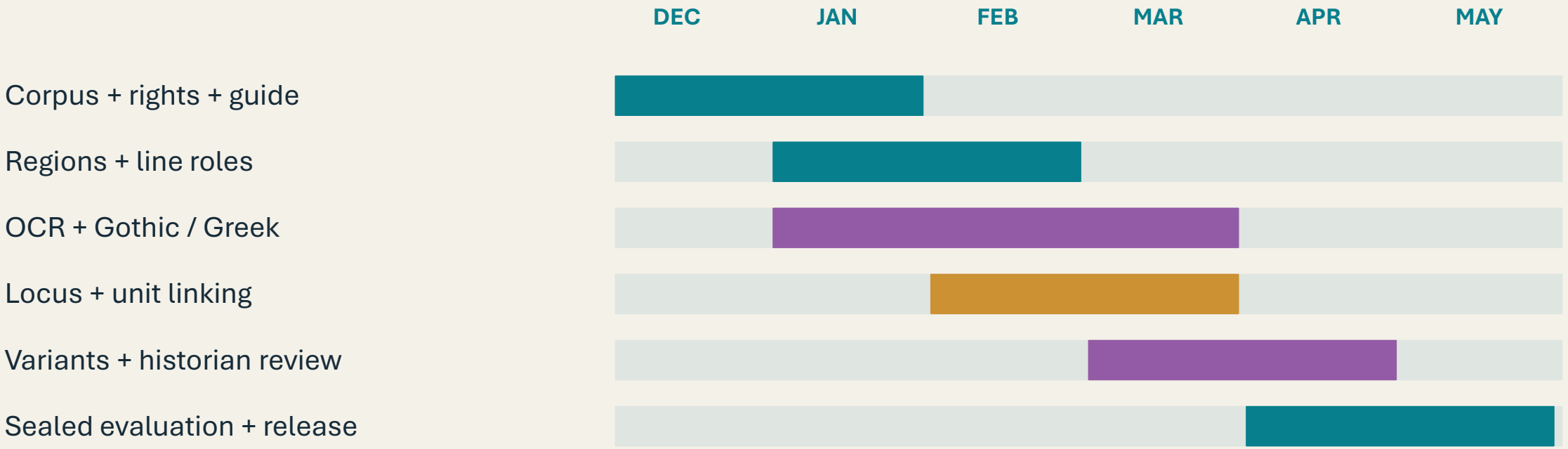
Names · citations · abbreviation readings
Attribution · conditions · exceptions

Draft → correct → discuss disagreements → update the guide

December: 12-page calibration workshop
Weekly: focused review clinic + annotated examples
Evaluation: double-review 60 focal units, then adjudicate

Six months, with a usable result at each stage

The existing software becomes the starting point for the pilot.



End January: checked pilot data
End March: linked paratexts

End May: tested variant workflow + first attribution-change study

Success means recovering the historical evidence

Evaluate on complete pages and matched units, then measure the cost of correction.

Regions + roles	Find full units, preserve streams	**Region F1 ≥ .90; role macro-F1 ≥ .95**
OCR + anchors	Recover the fields comparisons depend on	**Names / citations exact recovery ≥ 95%**
Linking + variants	Attach and group the correct paratexts	**Locus linking F1 ≥ .95; grouping F1 ≥ .90**
Substantive events	Detect attribution and qualification changes	**Precision ≥ 95%; recall ≥ 90%**
Historian effort	Reduce the work needed per usable unit	**Median correction time at least halved**

Pilot targets · report by edition, typeface and class · score all 60 sealed test pages

Publish a reusable corpus and an answerable question

The first study: when and where did named interpretation become anonymous?

GitHub

Versioned code · run scripts · tests
Annotation guide · evaluation protocol

Hugging Face / archival release

Model weights · permitted annotations
Dataset / model cards · versioned identifiers

Keep rights at the item level

Library identifiers + IIIF links + retrieval scripts
Redistribute images only under their recorded terms.

Matched loci × editions × attribution events

A corpus-wide pattern, with every conclusion linked back to its witnesses.

The committee can reproduce the result locally

See the package README for the supplied Windows launchers.

Minimal task

```
python -m pip install opencv-python numpy
python task1.py
```

MDZ interfaces

IIIF image API: api.digitale-sammlungen.de/iiif/image/v2/bsb10144443_00040
hOCR: api.digitale-sammlungen.de/ocr/bsb10144443/40
Image: use 00041. OCR: use /41 for the second task page.

Neural v3: the complete confusion matrix

20 evaluation pages · reference in rows, prediction in columns

	Text	Summarium	Inscriptio	Other	Outside
Text	**811**	0	0	1	0
Summarium	0	**143**	0	0	0
Inscriptio	4	1	**26**	0	0
Other	4	2	1	**36**	0
Outside	1	0	0	0	**4035**

Correct: 5,051 / 5,065 · Four-role macro-F1: 0.9450

The shipped models and evaluation sets

Distinct stages have distinct datasets and outputs.

D-FINE Large · epoch 14 / step 1400

100 training pages from 20 books; 20 evaluation pages from 20 other books.
One text-region class; 1,280-pixel input; score threshold 0.50.

PERO ParseNet_296000 + OCR_350000

Line detection and recognition run inside the D-FINE regions.
Language-model decoding is switched off.

Neural v3 · full.pt

100 training + 20 evaluation pages from ten Roman-glossed books.
Colour image + raw characters + coordinates + transformer page context.